Food Recognition and Nutritional Estimation Using Deep Learning

Deep Learning Final Project

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PyTorch • MobileNetV2

20-Class Food Image Classification



Problem & Objective

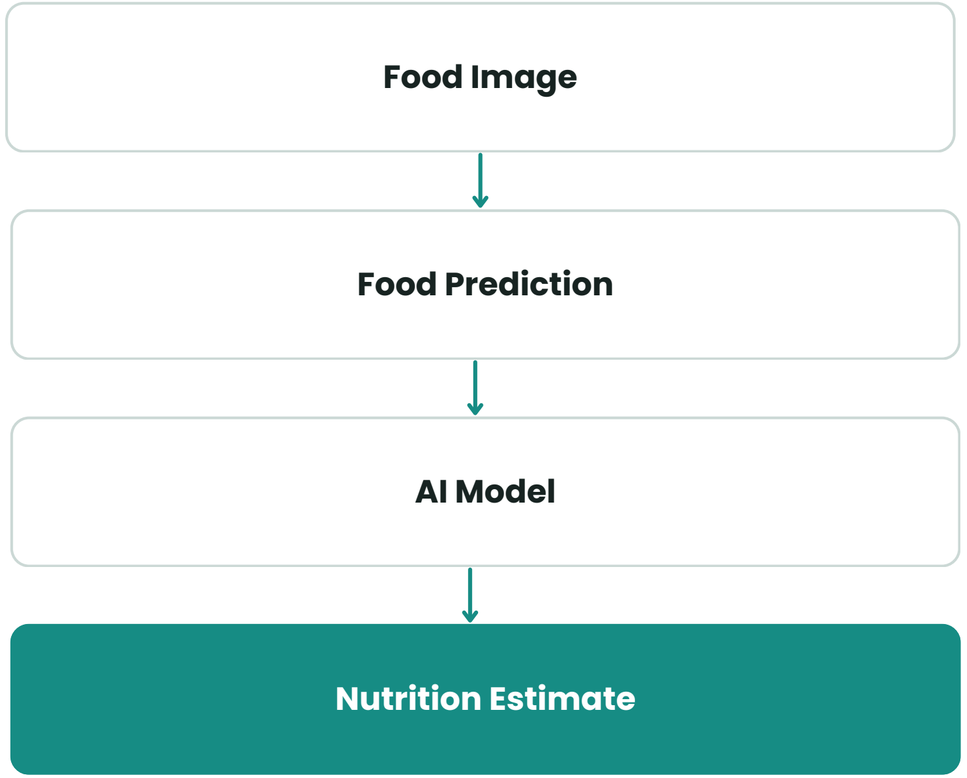
From a food image to a nutrition estimate

PROBLEM

Identifying food manually
can be slow and inaccurate.

OBJECTIVE

Classify images into
20 food classes and
estimate nutrition.



Dataset

3,058 images across 20 food classes and five categories.

2,151	596	311	3,058
Training	Validation	Test	Total

Category	Food Classes
Fast Food	Pizza, Hamburger, French Fries, Hot Dog
Fruit	Apple, Banana, Orange, Strawberry
Vegetable	Carrot, Broccoli, Tomato, Cucumber
Nuts	Almond, Peanut, Cashew, Walnut
Dessert	Donut, Ice Cream, Chocolate Cake, Cupcake

Data Preparation

A consistent 224 × 224 input pipeline for MobileNetV2.

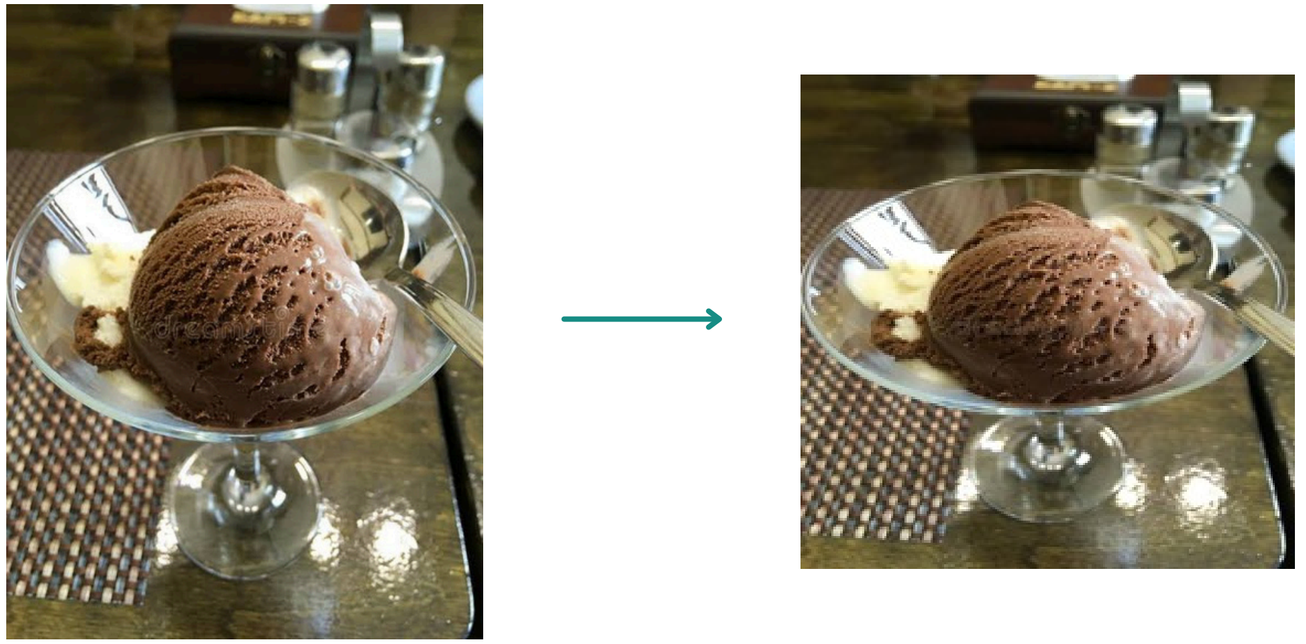
Food Image → Resize 224 × 224 → Augmentation (training only) → Tensor conversion → ImageNet Normalization → MobileNetV2 input

TRAINING

Horizontal flip + rotation up to 10°

VALIDATION / TEST

Resize + normalization only



Approach 1 — Frozen MobileNetV2

Use: Frozen MobileNetV2

Pretrained on ImageNet

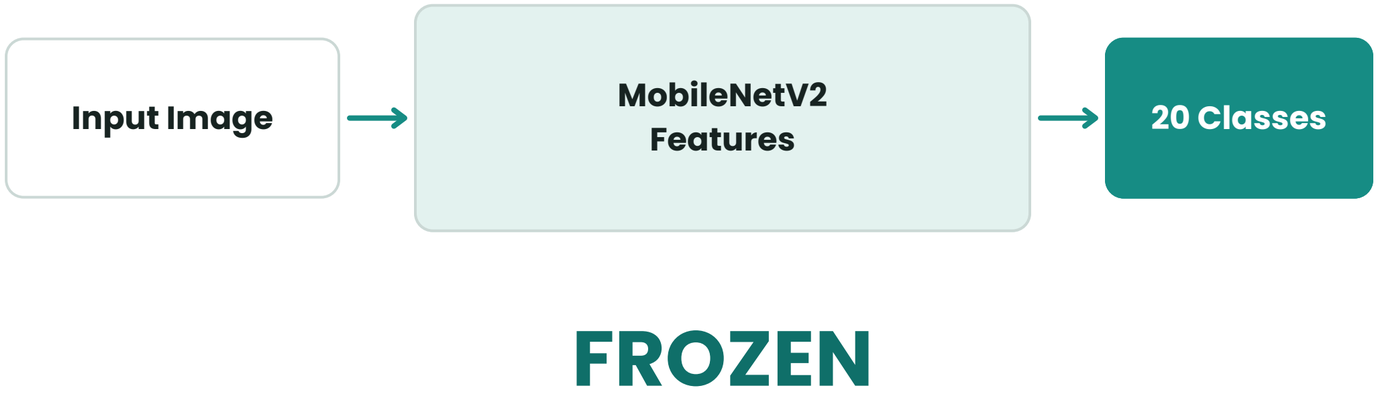
Feature backbone frozen

Only classifier trained

Epochs: 5

Learning rate: 0.001

Trainable parameters: 25,620



92.28%

TEST ACCURACY

Improving the Best Model

Targeted data improvement—not a third approach.

Before–95.18%

Improvement

+ diverse training images

6 weak classes (Apple, Carrot, Orange, Hamburger, Stawberry, Almonds)

LR: 0.00005

3 more epochs

After–95.50%

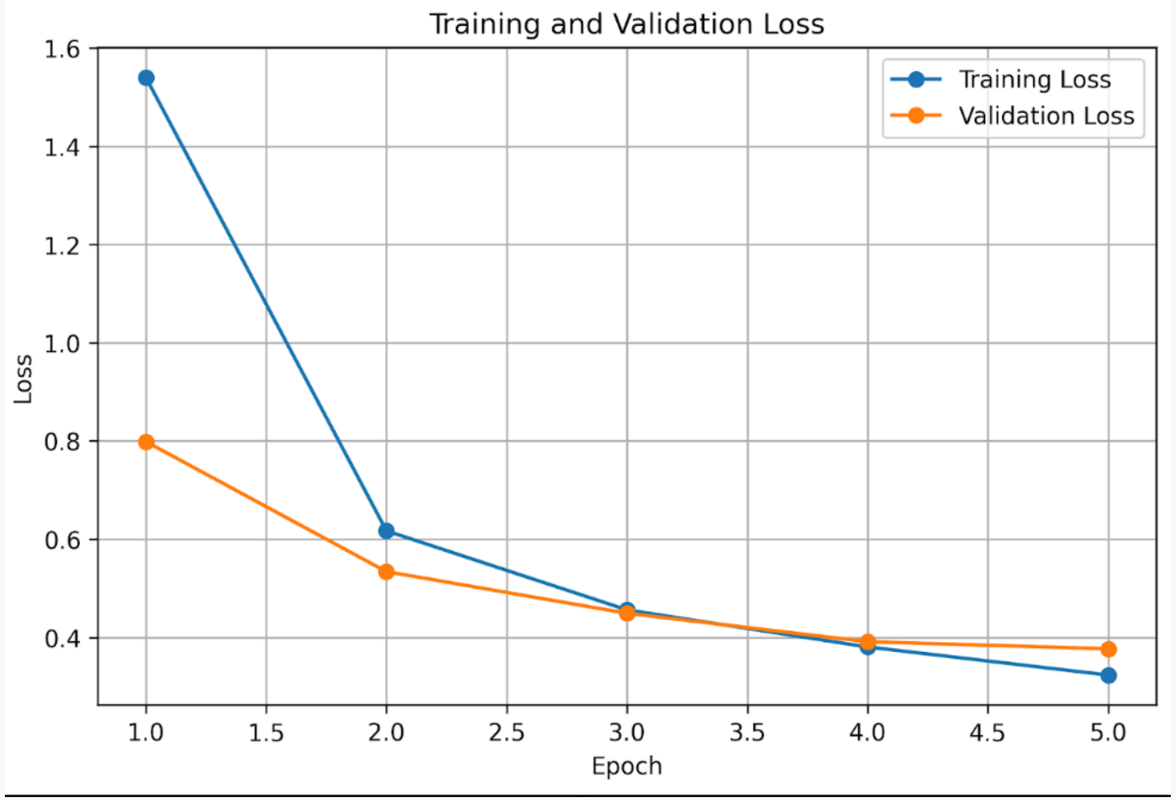
297 / 311 CORRECT

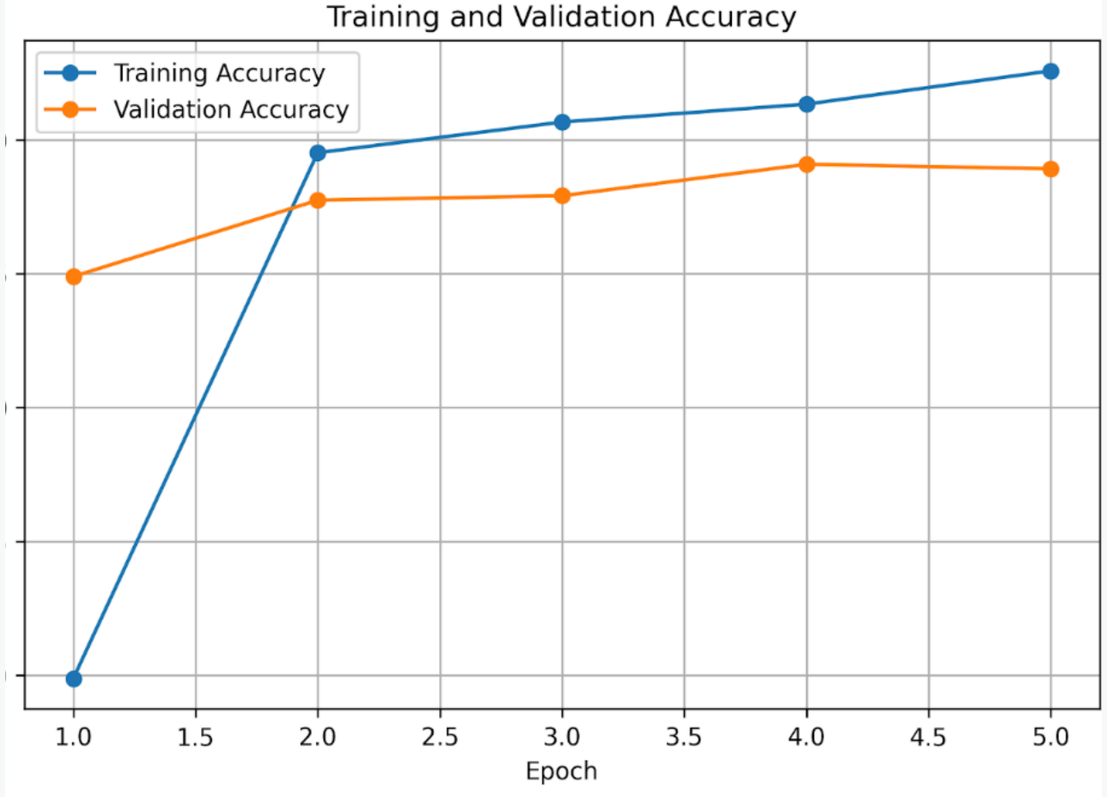
Experiment Setup

Framework	PyTorch
Architecture	MobileNetV2
Input / batch	224 × 224 / 32
Loss / optimizer	CrossEntropyLoss / Adam
Classes / test	20 / 311 images
Learning rates	0.001 / 0.0001 / 0.00005

Model Comparison

Fine-tuning delivered the largest gain; targeted data added a final lift.





System Demo & Nutrition Estimation



50% confidence threshold

Below 50% → “Uncertain Prediction”; no nutrition data

At or above 50% → food and nutrition displayed

Nutrition per 100g × portion grams / 100

Nutrition lookup is a feature—not a second deep learning model.

Conclusion & Future Work

KEY RESULTS

92.28% → 95.18% → 95.50%

- Frozen MobileNetV2: 92.28%
- Partial fine-tuning: 95.18%
- Final improved model: 95.50%

Limitations

- Only 20 food classes
- Unknown foods may be misclassified
- Portion size must be entered manually
- Limited real-world training images

Future Work

- Add more food classes
- Collect more real-world images
- Improve unknown-food detection
- Automatic portion estimation
- Compare other architectures

Fine-tuning and better training data improved the model from 92.28% to 95.50%.